

UNIVERSITI TUNKU ABDUL RAHMAN

EXAM SPECIMEN PAPER

UCCC 3073 / UCCG 3073 DATA SCIENCE

BACHELOR OF COMPUTER SCIENCE (HONS)
BACHELOR OF SCIENCE (HONS) STATISTICAL COMPUTING AND OPERATIONS
RESEARCH

Instruction to Candidates:

This question paper consists of **FIVE (5)** questions.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answers more than ONE (1) question in Section B, marks will only be awarded for the FIRST (1) question in that section in the order the candidate submits the answers.

Candidates are allowed to use any type of scientific calculator.

Answer questions only in the answer booklet provided.

UCCC 3073 / UCCG 3073 DATA SCIENCE**SECTION A (Answer ALL questions)**

Q1. (a) Briefly discuss the business understanding stage in the data science methodology and TWO (2) reasons why it is important. (6 marks)

(b) Big Data is often referred to as having certain attributes or characteristics. List and describe any TWO (2) of these characteristics. (6 marks)

(c) Consider the data set below. State whether each attribute contains numeric, interval, ordinal, ratio, nominal, categorical. (5 marks)

Sepal length	Sepal width	Petal length	Petal width	Species
6.2	≤ 3	Moderate	1.3	Iris-versicolor
5.1	≤ 3	Small	1.1	Iris-Versicolor
6.3	> 3	Big	2.5	Iris-virginica
10.8	< 3	Big	1.9	Iris-virginica

(d) Normalise the following data using the given methods:

200, 300, 400, 600, 800, 1000

(i) A standard min-max normalization. (2 marks)

(ii) z-score normalization. (2 marks)

$$x' = \frac{x - \bar{x}}{\sigma}, \text{ where } \bar{x} = \frac{\sum x}{n}, \sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

(e) Partition the following data into three bins using the given methods:

15, 10, 11, 250, 35, 50, 55, 92, 85, 88, 169, 204

(i) Equal-depth (frequency) partition. (2 marks)

(ii) Equal-width (distance) partition. (2 marks)

[Total : 25 marks]

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Q2. (a) Given the following data set of Mr. N's purchasing history in Google Play Store:

App ID	Price	Type	Review	Purchase?
1	1.99	Game	Average	Yes
2	4.99	Productivity	Good	No
3	2.50	Camera	Average	No
4	5.90	Game	Excellent	Yes
5	4.20	Camera	Excellent	No
6	3.49	Game	Good	No

Table Q2a

- (i) Draw a table to identify frequency counts and probabilities for each attribute in Table Q2a. (5 marks)
- (ii) Using the frequency table you have produced in Q2(a)(i), apply Bayes' rule of conditional probability to calculate the probability that the Mr. N will make the purchase. (5 marks)

App ID	Price	Type	Review	Purchase?
7	4.99	Game	Excellent	?

- (b) Given the training data (0.9, 2.1), (0.5, 1.1), (0.65, 1.5), (0.825, 1.9). You believe that the data should fit a linear function. Determine the following:
- (i) the best-fit regression line for the data. (8 marks)
- (ii) the value of y predicted when the input x is 1.5. (2 marks)
- (iii) the R^2 coefficient. (5 marks)
- [Total : 25 marks]

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Q3. (a) Given the following is the transaction records of a customer's purchases:

17 Sep 1960	24 Sep 1960	1 Oct 1960	8 Oct 1960
- Anvil - TNT	- Boomerang - Carrots (Iron) - TNT	- Dehydrated Boulders - Earthquake Pills - TNT	- Earthquake Pills - TNT

- (i) Describe the technique, 'association rules mining'. (2 marks)
 - (ii) Use the Apriori algorithm to find all the itemsets with support threshold of 2. Show the candidate itemsets obtained at each stage of the algorithm with their support. (6 marks)
 - (iii) Name TWO (2) interesting trends which you can observe from the transactions. (2 marks)
 - (iv) Name TWO (2) kinds of knowledge, other than frequent itemsets, that supermarkets look for in transaction data. (2 marks)
 - (v) Describe any issues or limitation of association rules mining. (3 marks)
- (b) The following graph plots two groups of data, marked with respectively X and o. We are interested in whether a new data point () should belong to first or the second group of data.

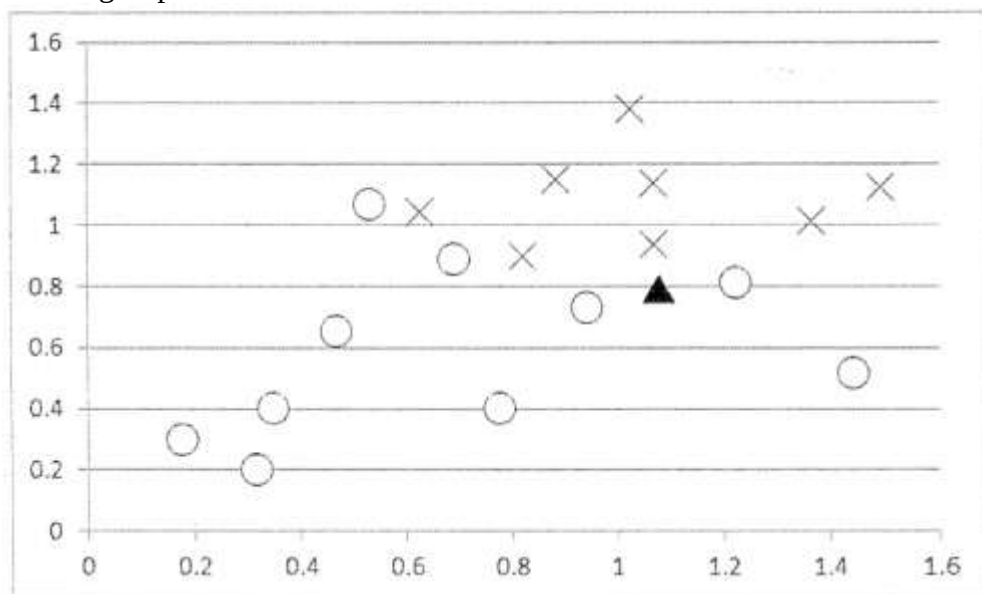


Figure Q3(b)

- (i) Determine the class for the new data point if 1-NN classification is used. (2 marks)

[Q3 continue]

- (ii) Determine the class for the new data point if 3-NN classification is used. (2 marks)
 - (iv) Explain how an increase in k would affect the expected loss in a k -NN classification in terms of Bias, Variance and Noise. (6 marks)
- [Total : 25 marks]

SECTION B (Choose only ONE question)

- Q4. (a) In a given classification task, the examples were split into a training set and a validation set. Figure Q4(a) shows the classification performance of the classifier on example sets of various sizes from the training set. One of the lines corresponds to the error rates when tested on the training set, while the other line corresponds to the error rates when tested on the validation set.

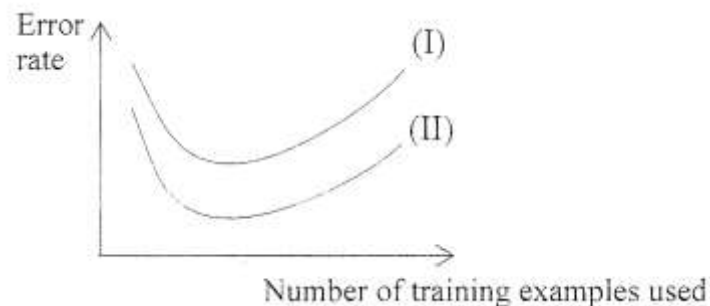


Figure Q4(a)

- (i) Which of the line is more likely corresponds to the error from the validation set. Justify your answer. (3 marks)
- (ii) Justify when should the training be stopped. (2 marks)
- (iii) Determine if the current learning model suffers from over-fitting or under-fitting. (3 marks)
- (iv) Determine which error rates should be used when reporting the error rate of the trained learner. (2 marks)

[Q4 continue]

- (b) The table below shows the predictions made by a Naïve Bayes classifier to classify dogs.

ID	Target	Prediction
1	dog	dog
2	cat	cat
3	cat	cat
4	cat	cat
5	dog	dog
6	cat	dog
7	dog	dog
8	dog	dog
9	cat	cat
10	cat	cat

Calculate the evaluation measures as following:

- (i) A confusion matrix and the misclassification rate. (4 marks)
- (ii) The precision, recall, and F_1 measure. (6 marks)
- (c) The following formula shows that accuracy is a function of sensitivity and specificity.

$$accuracy = sensitivity \cdot \frac{P}{P + N} + specificity \cdot \frac{N}{P + N}$$

Prove this equation.

(5 marks)

[Total : 25 marks]

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Q5. Given a mail marketing campaign on customers response to the newly launched product. The decision tree algorithm is used to predict customer respond to the campaign. Category 1 indicate positive response and category 0 indicate negative response:

(a) Given the following is the confusion matrix of the model:

Model 1 prediction with equal cost			
N = 10,000		0	1
Actual	0	1010	1990
values	1	2200	4800

Table Q4a

- (i) Calculate the baseline performance. (2marks)
- (ii) Calculate misclassification rate, sensitivity, specificity and F_1 score. (4 marks)
- (iii) Justify whether or not the model is fit for the data. (5 marks)
- (b) Assume that unequal cost on false positive (FP) and true positive (TP) are implemented in the model, the following is the confusion matrix of the model when cost for false positive, $Cost_{FP}$ is 10 and cost for true positive, $Cost_{TP}$ is – 40.

Model 1 prediction with unequal cost			
N = 10,000		0	1
Actual	0	1350	1650
values	1	500	6500

Table Q4b

- (i) Calculate misclassification rate, sensitivity, specificity and F_1 measure. (4 marks)
- (ii) Calculate the model cost per record (3 marks)
- (iii) Justify whether or not the unequal cost model is better than the baseline performance and the equal cost model in Q4a. (7 marks)
- [Total : 25 marks]

APPENDIX

Regression	
Straight line, $\hat{y} = \beta_0 + \beta_1 x$	t value, $t = \frac{\hat{\beta}}{SE}$
Coefficients $\hat{\beta}_1 = \frac{p\sum xy - \sum x \sum y}{p\sum x^2 - (\sum x)^2}$	Standard errors of coefficients $SE(\hat{\beta}_1) = \frac{s}{\sqrt{\sum (x - \bar{x})^2}}$
$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$,	$SE(\hat{\beta}_0) = s \sqrt{\frac{1}{p} + \frac{\bar{x}^2}{\sum (x - \bar{x})^2}}$
Coefficient of Determination $R^2 = \left[\frac{p\sum xy - \sum x \sum y}{\sqrt{p\sum x^2 - (\sum x)^2 * p\sum y^2 - (\sum y)^2}} \right]^2$	Residuals $\hat{\varepsilon} = y - \hat{y}$
$R = \sqrt{R^2}$	$s^2 = \frac{\sum \hat{\varepsilon}^2}{p - 2}$
	$s = \sqrt{s^2}$

Naïve Bayes
Bayes' rule, $P(H E) = \frac{P(E H)P(H)}{P(E)}$
Laplace correction, $\frac{x_i+1}{c+\alpha}$
Probability density function for a Gaussian distribution, $pdf(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\bar{x})^2}{2\sigma^2}}$

Association Rules
Support, $s = \frac{freq(X,Y)}{N}$
Confidence, $c = \frac{freq(X,Y)}{freq(X)}$
Lift, $l = \frac{Support}{Supp(X)*Supp(Y)}$

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Evaluation metrics	
Classification accuracy, $acc = \frac{TN+TP}{TN+TP+FN+FP}$	Precision, $PPV = \frac{TP}{TP+FP} = 1 - FDR$
Misclassification rate, $1 - acc$	F1 score, $F_1 = \frac{2TP}{2TP+FP+FN}$
Sensitivity / Recall, $recall = \frac{TP}{TP+FN}$	Overall model cost $= TN.Cost_{TN} + FP.Cost_{FP} + FN.Cost_{FN} + TP.Cost_{TP}$
Specificity, $spec = \frac{TN}{TN+FP}$	Model cost per case $= \frac{Overall\ model\ cost}{N}$
False discovery rate, $FDR = \frac{FP}{FP+TP}$	Model profit per case $= -model\ cost\ per\ case$

Proximity measures	
Euclidean distance, $d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$	Hamming distance, $d(a, b) = (a \oplus b)$
Manhattan distance, $d(x, y) = \sum_{i=1}^n x_i - y_i $	Simple matching coefficient, $SMC = \frac{num\ of\ matching\ attribute\ values}{num\ of\ attributes}$
Supremum distance, $d(x, y) = \max(x_i - y_i)$	Jaccard coefficient, $J = \frac{num\ of\ matching\ presences}{num\ of\ attributes\ not\ involved\ in\ 00\ m}$
	Cosine similarity, $cos(x, y) = \frac{x \cdot y}{ x y }$

Miscellaneous	
Min-Max normalisation, $x' = \frac{x - \min(x)}{\max(x) - \min(x)}$	Confidence intervals, $CI = \bar{x} \pm z * SE$
z-score normalisation, $x' = \frac{x - \mu}{\sigma}$	Standard error, $SE = \frac{\sigma}{\sqrt{n}}$
Sample mean, $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$	Z-test, $Z = \frac{\bar{x} - \mu}{SE_{\bar{x}}}$
Sample standard deviation, $s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$	